

1.1 Defined and Undefined Terms – Part 1

Lesson Introduction				
 Benchmark	MA.912.GR.2.2 Identify transformations that do or do not preserve distance.		 Learning Targets	- I can use rigid motions to define parallel lines and perpendicular lines. - I can experiment with rigid motions on a plane. - I can use rigid motions to define a right angle.
 Benchmark Coherence	Building On		Working Toward	
	8.GR.2.1 Given a preimage and image generated by a single transformation, identify the transformation that describes the relationship.		N/A	
 Essential Questions	- What is a line? - How do you label a line and a line segment? - What is a plane? - What are parallel lines? - What are perpendicular lines?			
 Lesson Narrative	This is the first part of a two-part lesson. Students use patty paper to explore facts about lines and planes, developing an essential foundation of motions on a plane that will be used throughout the course of Geometry. Students will engage in collaboration about how motions on a plane can be used to formulate definitions by observing key details of transformations. The lesson applies transformations to help students explore important terms and relationships that are foundational.			
 Component Summary	1.1.1 Warm-Up (~5 min.)	Estimate location of one billion on a number line (<i>SE p. 3</i>)	1.1.3 Collaboration (15 min.)	Define key foundational words of Geometry through motions on a plane (<i>SE p. 7</i>)
	1.1.2 Guided Instruction (25 min.)	Define key foundational words of Geometry through motions on a plane (<i>SE p. 4</i>)	1.1.4 Wrap-Up (~5 min.)	Reflection rating of lesson understanding (<i>SE p. 9</i>)
 Resources	Glossary		Materials	
	<u>Key Terms:</u> - Line - Line Segment - Plane - Parallel Lines - Right Angle - Perpendicular Lines - Skew Lines <u>Additional Terms:</u> - Reflection		- Patty Paper (Optional) Protractors (Optional) Card Stock (Optional) Sticky Notes	
			- Consider reviewing Jo Boaler’s research on how mistakes grow the brain. - Complete 1.1.2 Guided Instruction question 3 before class, saving each step on a separate piece of patty paper so you can quickly display it if needed. - Honors Geometry students may be able to complete lessons 1 and 2 together. - Create sentence stems that students can use to discuss strategies in 1.1.1 Warm-Up.	

 Teacher Tips	Common Misconceptions	Things to Consider
	- Students may struggle with the difference between a line and a plane. - Students may have a difficult time understanding that a line continues forever in both directions. - Some may have a difficult time visualizing a plane.	- Before starting 1.1.1 Warm-Up, consider putting a blank number line on a wall, possibly with chart paper or by projecting it. After students have completed the 1.1.1 Warm-Up, have a few students place sticky notes where they believe the number one billion is. Students may not be familiar with each other (or with you). Consider utilizing the 1.1.1 Warm-Up as an opportunity to establish, discuss, and practice classroom norms. - This lesson could easily be divided up in different ways so that while establishing class norms for your class students are also putting these norms into practice. - Creating a Word Wall may be helpful at the beginning of this unit. This will set a positive tone for creating Word Walls throughout the course.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Stronger and Clearer Each Time Starting with the first lesson of the course, students have many opportunities to work with a partner to help define terms and ideas. This Literacy Routine allows students to refine and clarify thoughts they have, allowing them to end the lesson with a solid grasp on the lesson content.	MA.K12.MTR.4.1: Discussion and Reflection Students will be working with partners and other pairs to develop and fine-tune definitions of key words. In 1.1.2 Guided Instruction, students write definitions, then are prompted to collaborate with classmates to fine-tune and verify their definition.
 Differentiate Instruction	For 1.1.2 Guided Instruction and 1.1.3 Collaboration, consider providing the copied images and partial steps on sheets of patty paper for those students who struggle with the physical dexterity needed to trace and fold the papers. This step will allow the students to make observations of the motions without hindrance.	
Lesson Notes:		

1.1.1 Warm-Up: A Billion!

Component Overview: Students estimate the location of 1 billion on a given number line and describe the strategy they used to determine the location.



1.1.1 Warm-Up: A Billion!

One billion is a big number. Starting the school year, it may feel like there are a billion seconds left in the year. When you started school there were 1.555×10^7 , or 15,550,000 seconds left in the school year.

A billion seconds is approximately 32 years!

1000000000	=	31.7097919838
Second		Calendar year

So, although it may feel like a billion, it really isn't.

1. What is the location of 1 billion on the line segment?



2. Describe your strategy for estimating the location of 1 billion on the line segment.

Answers will vary. I approached this question thinking that a billion was very close to the trillion mark. After thinking about the number of zeros, a trillion has three more zeros, making it a thousand times bigger than one billion. So my mark represents one thousandth of the way from zero to one trillion.



The movie “Paper Clips” deals with this very idea of relating large numbers to students. Showing or discussing this film may be a great way to connect with Social Studies or English and help students begin to develop the idea of how large numbers really are. Specifically, the movie is about a project that tried to illustrate to students the magnitude of the 6 million Jews killed during the Holocaust.



Consider using 1.1.1 Warm-Up to establish class norms on working with partners by pairing students to discuss their strategies. Establish the concept of mistakes as learning opportunities by using positive statements about the mistakes and how mistakes are an opportunity to learn.



If debriefing this activity prior to moving to 1.1.2 Guided Instruction, consider these questions:

- What value would be located in the middle of this line segment?
- How could this line segment be divided into smaller intervals?
- How many times smaller is 1 billion than 1 trillion?

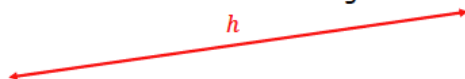
1.1.2 Guided Instruction: Definitions

Component Overview: After writing their best definition of a line segment and line, students are presented with facts about lines and planes. Using patty paper, students explore how motions on a plane are used to develop definitions.



1.1.2 Guided Instruction: Definitions

1. In the Warm-Up, the diagram is a line segment.
 - a. Discuss with your partner why it wasn't called a line.
Answers will vary. If a movie segment is a small "part" of a movie, then a line segment must be just a small part of a line.
 - b. Write your best definition of a line segment.
Answers will vary. A line segment is part of a line that is clearly beginning and ending at exact points.
 - c. Draw a line. Label the line using the letter h .



- d. Write your best definition of a line.
Answers will vary. A line is a way of connecting a bunch of points that continue in both directions for ever.
- e. Ask another partner pair the definitions they wrote. Record their definition and write your summary of their reason for the definition. *You are the only person who should write in the first two columns of the table.* Have the person initial next to your summary stating that your summary was correct.
Answers will vary.

Definition	My Summary of Their Reason	Initials
line segment		
line		



Question 1a is a discussion opportunity. Students do not need to write an explanation here, but a sample response is provided for reference. Listen as students discuss their thinking in response to this prompt to determine student thinking to highlight or misconceptions to address. 1.1.2 Guided Instruction is scaffolded to allow students to engage with content with their partner, the class, and with the teacher. Establishing an atmosphere where students may revise their thinking throughout instruction as they engage and discuss provides a safe atmosphere for students to explore and to be wrong.



Students are being asked to write their best definition of a line and line segment. The intent here is not to provide the definition for them and teachers should avoid any clarification of their definitions. In Lesson 2 of Unit 1, the "undefined terms" of Geometry will be introduced — point, line, and plane.

1.1.2 Guided Instruction: Definitions (continued)



Facts About Lines

- A line has infinite length, with no height and no width.
- A line is assumed to be straight.
- A line is considered one-dimensional.
- A line is considered an undefined term.
- A line goes on forever in both directions.



While discussing each of these facts about lines, avoid spending too much time on bullet 4, as an “undefined term” will be clarified in the next lesson. Focus more on the other four bullets and what they mean to the students.

A line has infinite length, thus, any line drawn should have arrowheads on both ends.

A line can be named by a single lowercase script letter, or by any two (or more) points which lie on the line.

2. Consider the two lines shown.



- Use the script lowercase letter m to name the purple line.
 $\overleftrightarrow{m}$
- Use the letters H and R to name the orange line.
 $\overleftrightarrow{HR}$



Consider these discussion prompts after question 2:

- What does “infinite length” mean?
- How many points are on a line?



Facts About Planes

- A plane has infinite length and width, with no height.
- A plane is considered two-dimensional.
- A plane is considered an undefined term.
- A plane goes on forever.

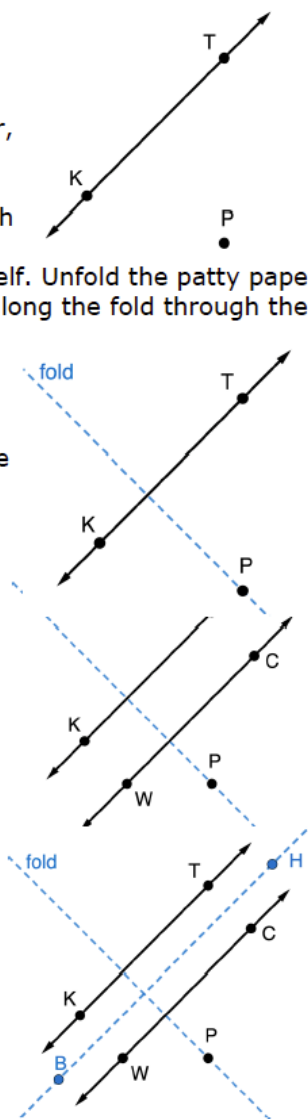
1.1.2 Guided Instruction: Definitions (continued)

A plane is named by a single uppercase letter written in script or by three points that do not lie on the same line.

3. Consider the line and point shown.

a. Complete the following:

- On a piece of patty paper, copy line $\overleftrightarrow{KT}$ and point P .
- Fold the patty paper so that the fold goes through point P and each part of line $\overleftrightarrow{KT}$ coincides with itself. Unfold the patty paper and draw a dashed line along the fold through the point, as shown below.
- Fold the dotted line on itself so that point P coincides with $\overleftrightarrow{KT}$.
- The resulting fold is a line that is parallel to the $\overleftrightarrow{KT}$. Use a writing utensil and straightedge to draw the parallel line. Label the line using letters W and C as shown below.
- Show the lines are parallel by folding the patty paper so that $\overleftrightarrow{KT}$ and $\overleftrightarrow{WC}$ coincide. Unfold the paper and draw a dashed line along the fold and name it $\overleftrightarrow{BH}$.



It may be important to read the directions here, replacing the word “coincides” with the words “lines up” or “match” to help students remember what the word “coincides” means.

1.1.2 Guided Instruction: Definitions (continued)

- b. Complete the statement.

$\overleftrightarrow{WC}$ is a
☐ rotation
☒ reflection
 of $\overleftrightarrow{KT}$.



The focus of this lesson is not on transformations. Avoid spending more than a few moments reviewing rotations and reflections, as they will be further explored in future lessons.

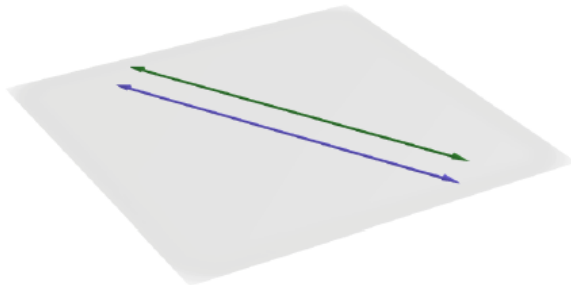
- c. Work with your partner to describe how the lines you drew on patty paper, $\overleftrightarrow{KT}$ and $\overleftrightarrow{WC}$, lie on the same plane.

Answers will vary. The patty paper represents a plane. Since I can draw $\overleftrightarrow{KT}$ and $\overleftrightarrow{WC}$ on the paper, then they are on the same plane.



Parallel lines do not intersect and lie in the same plane.

The diagram shows two parallel lines that lie on a plane. The plane is represented by the gray rectangle.



1.1.3 Collaboration: Definitions

Component Overview: Students continue using the same patty paper to explore the definition of perpendicular lines.

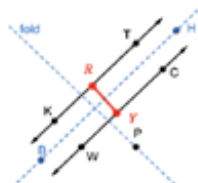


1.1.3 Collaboration: Definitions

- Continue using the patty paper from the previous component.

- Label a point on $\overleftrightarrow{KT}$ as R . Then, fold the patty paper along $\overleftrightarrow{BH}$ and copy point R to $\overleftrightarrow{WC}$ naming it Y . Unfold the paper and draw a line segment that connects points R and Y .

Example shown.



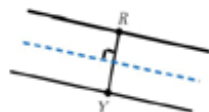
- What type of angle does $\overline{RY}$ form with the dashed line?

The line crosses at a right angle.



A **right angle** is an angle measuring exactly 90° .

- Use a ruler to measure the distance between point R and the dashed line, and between point Y and the dashed line. Write the measurements, with units, on the diagram.



Answers will vary depending on the units. The distances between point R and the dashed line and point Y and the dashed line are the same, 1 cm each.

- Discuss with your partner: when you reflect an object across a line, will corresponding points be the same distance from the line of reflection? Summarize your discussion.

Answers will vary. Yes, a true reflection means that all parts are the same distance, so the distance something is from the line of reflection must also be the same for the reflected item.



Students may not have ever used patty paper before. Allow students the space to make mistakes and to start over. As you circulate be prepared to help students who are struggling with using the patty paper.



Provide protractors for students to verify that the angle measures 90° . Or suggest to students to use a corner of a sheet of paper to verify the angle is a right angle.

1.1.3 Collaboration: Definitions (continued)

2. Complete the statements.

The patty paper is similar to a ☐ plane.
☐ reflection.

By reflecting a line across a line of ☐ reflection,
☐ rotation,

a parallel line was created. The parallel line ☐ is
☐ is not

the same distance from the line of ☐ reflection.
☐ rotation.

3. How is $\overline{RY}$ related to the dashed line?

$\overline{RY}$ is perpendicular to the dashed line because it is crossing the dashed line at exactly 90° .



After questions 1c and 1d, consider asking if this constant distance between parallel lines is important in the real world. Possibly in architecture, railroads, or other applications.



Perpendicular lines are lines that intersect at right angles.

The table shows multiple examples of perpendicular lines.

Description	Example of Perpendicular Lines
Perpendicular lines $\overline{KT}$ and $\overline{RB}$ shown in plane X .	
Perpendicular lines $\overline{KT}$ and $\overline{RB}$ shown in planes M and W .	
Perpendicular lines $\overline{KT}$ and $\overline{RB}$ shown in planes M , W and X .	

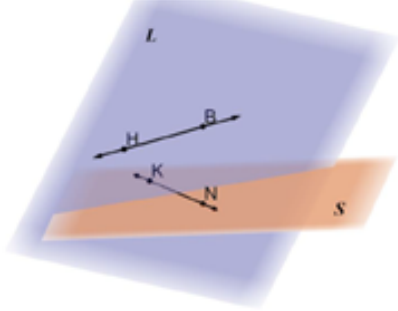
1.1.3 Collaboration: Definitions (continued)

4. If two lines are perpendicular, then any two adjacent angles formed are right angles.



Skew lines are lines that do not intersect and are not parallel.

The table shows an example of skew lines.

Description	Example of Skew Lines
Skew lines $\overleftrightarrow{HB}$ and $\overleftrightarrow{KN}$ shown in planes L and S .	



Consider asking students what they notice about the intersection of two planes. Or consider asking them to complete the statements:

Two lines in a plane intersect at _____. (a point)
The intersection of two planes is a _____. (a line)



To further help students visualize planes use a sheet of paper to represent a plane and two sheets of paper, where one has a slit, to show the two planes. Card stock or cardboard may also be helpful.

1.1.4 Wrap-Up: Reflection

Component Overview: Students self-reflect on the lesson and provide numerical values to rate their understanding.



1.1.4 Wrap-Up: Reflection

1. Rate yourself based on what you understood in today's lesson.

Answers will vary.

Skill	1 – I am just starting. 2 – I have some skills. 3 – I am almost there. 4 – I have it!
I can define parallel lines.	
I can define perpendicular lines.	
I can define a right angle.	

2. Rate yourself based on how you feel about the following statements.

Answers will vary.

Statement	1 – I struggle with this. 2 – I am ready to grow in this area. 3 – I am almost there. 4 – That's me!
I am willing to try new things.	
I persevere when something is challenging.	



For students who respond with many threes or fours, prompt them to discuss applications for each of the key words found throughout the lesson. Is there a real-world application, or a mathematical application? Having students "predict" test questions would also be effective.



Consider our online resources for additional enrichment opportunities for students to practice or test their abilities.